

Flutter and Dart Training Documentation

Topic: Dart and Flutter

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1. Dart

1.1 . Introduction to Dart

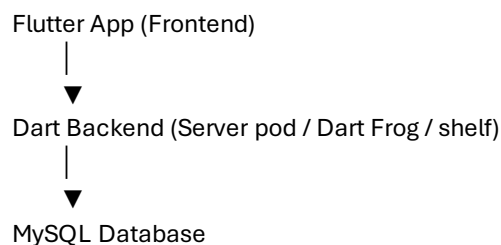
1.1.1. What is Dart? - Dart is an open-source, object-oriented, **statically typed** programming language developed by google. So, it is mainly used with the flutter framework to build applications for Android, iOS, Web, Desktop, and Backend from a single codebase.

Flutter – is a framework used to build UI. So, we need a programming language to build the Application Logic, calculation, talk with API, and manage Application data. So, we use darts.

Dart – is a programming language used to write all code including

- Business logic
- State management
- API calls
- Navigation
- Validation

But we can also use the Darts for backend, but we are not using flutter for the backend. Dart can be used independently on server to build the backend Applications using the framework like Server pod, Dart Frog, or shelf.



Important Terms -

1. Open source – means the source code is available Publicly. So, anyone can use it, study it, and contribute to its development without paying any license fees.

No need to pay any license fee to contribute for its development Because it's an open-source Language.

2. Object-Oriented Programming (OOP) - so dart follows object-oriented programming.

Means everything is organized into classes and objects.

Ex like a Car is a class and the individually car AUDI, BMW, TATA, Mahindra is an object of car class. This makes programs easier to organize, reuse, and maintain.

3. Statically Typed Language – means this language is checking the data type of variable before the program runs.

Means it checks the data Type of variable at compile time this is called Statically Typed Language.

Ex: Dart, Java, C++

And dynamically typed language means – it checks the data type of variable at run time

Ex: language – Python, JavaScript

Dart Ex – `int age = 20;`

Age = “Atharv” - so it will give error

4. Cross Platform development - means writing a single codebase that can run on multiple OS and devices without rewriting the application for each platform.

Like Traditional development – Each Platform has its own language and framework like android has java/Kotlin, IOS has Swift, web have HTML, CSS and JavaScript.

Here only one project is created -

The **Flutter SDK** uses the **Dart compiler** to compile your application. So the process is -

You Write Code (Dart)

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Flutter SDK

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Uses

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Dart Compiler

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Native Machine Code

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Android / iPhone / Desktop executes it

Where is Dart Used? -

1. Mobile Applications -

- Banking apps
- Food delivery apps
- Shopping apps
- Chat applications

2. web Applications

- Admin dashboards
- Business management systems
- Company portals

3. Web Applications

Applications for:

- Windows
- macOS
- Linux

4. Backend Development

Dart can also create:

- REST APIs
- Web servers
- Backend services

Real World use-cases -

Banking app – secure Applications for money transfer.

Food Deliver app – Applications like Swiggy and Zomato.

Hospital app -Managing patients, doctors, and appointments.

1.1.2 Importance of Dart

It allows developers to build fast, Scalable and cross-platform applications from single codebase.

Flutter relies on dart because it provides a feature that helps developers to write code once and deploy on every platform.

1.single codebase for multiple platforms – real world example – suppose we are building a food delivery app

So, without dart: Android – Kotlin

And for IOS – swift

With dart – a single Dart project will run on any platform.

2. High Performance – dart application is so fast because it uses a two compilation Techniques during development.

1. JIT (Just-In-Time Compilation)

- Faster development
- Supports Hot Reload
- Code changes appear instantly

2. During Release – At the time of release we use a AOT compiler

AOT (Ahead-of-Time Compilation)

- Converts Dart code into native machine code before the app runs
- Better performance
- Smooth animations

Real world Ex –

When developing a shopping app - during coding and changes, we use JIT to test quickly. When publishing to the Play Store → AOT makes the app faster for users.

3. Hot reload – when we want a change on the screen and we want to inject our code into running application and want to rebuild the only that changed widget so that time we use Hot reload.

Instead of rebuilding the entire app flutter only rebuild the widget that been modified

Without Hot Reload -

We need to

1. change the code - suppose we changed the code
2. Stop the application – so now for compile we need to stop the Application
3. Compile again – after compile
4. Launch the Application – start the app and launch the app

5. Navigate back to the original screen - it shows by default its first screen, so we need to navigate to original screen.
6. Check the result

How Host reload works -

Write Dart Code

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Save File (Ctrl + S)

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Flutter Detects Changes

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Updated Code is Injected

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Widgets are Rebuilt

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UI Updates Instantly

